09

Load Balancing & Autoscaling

Spreading traffic across healthy backends and growing or shrinking capacity automatically

PCA study series · 11 concept slides · 8 scenario questions

Three load balancer families, each internal or external

1 Application Load Balancer

Layer 7 proxy for HTTP and HTTPS. Routes on host and path, terminates TLS, integrates with Cloud CDN and Cloud Armor.

2 Proxy Network Load Balancer

Layer 4 proxy for TCP, with optional SSL offload. Use for non-HTTP TCP services that benefit from proxying.

3 Passthrough Network Load Balancer

Layer 4 without a proxy: packets reach backends unchanged, client IP preserved. TCP, UDP, ESP, GRE, ICMP.

EXAM LENS Decide on three axes: HTTP or not, internet-facing or internal, global or regional.

Building a managed instance group

1

Instance template

Machine type, image, disks, metadata and startup script. Templates are immutable — create a new one to change.

2

Instance group

Zonal, or regional across up to three zones for resilience. Identical VMs created from the template.

3

Autoscaler

Policy that adds or removes VMs based on a signal, with minimum and maximum sizes.

4

Health check

Autohealing recreates VMs that fail the check; the initial delay protects slow-booting apps.

5

Updates

Rolling or canary updates move the group to a new template with control over surge and unavailability.

EXAM LENS Prefer regional MIGs: a zone outage removes only part of the capacity.

What the autoscaler can follow

1 CPU utilisation

The default. Set a target such as 60% so there's headroom while new VMs boot.

2 Load balancer serving capacity

Scale on utilisation or requests per second as reported by the load balancer's backend service.

3 Cloud Monitoring metrics

Any standard or custom metric, such as Pub/Sub backlog per VM or queue length.

4 Schedules and prediction

Scale ahead of known peaks, or let predictive autoscaling learn daily cycles.

EXAM LENS The initialisation period stops the autoscaler from judging a VM before its app is ready.

Two jobs for health checks

Load-balancer health check

Decides where traffic goes

- Unhealthy backends stop receiving requests
- Can be fairly aggressive to protect users
- Firewall must allow probes from 35.191.0.0/16 and 130.211.0.0/22

Autohealing health check

Decides whether to recreate a VM

- Failing VMs are deleted and replaced
- Use more conservative thresholds than the load balancer
- Set an initial delay longer than app start-up time

EXAM LENS 'Every backend shows unhealthy but the app works' → firewall rule for Google's probe ranges is missing.

Stateful managed instance groups

1 Why they exist

Some workloads (databases, legacy apps, Kafka brokers) need their disks, names or IPs to survive VM recreation.

2 What is preserved

Persistent disks, instance metadata, and internal or external IP addresses.

3 Still managed

Autohealing and rolling updates keep working, re-attaching the preserved state to replacement VMs.

4 Trade-off

Stateful groups can't use autoscaling, so pair them with planned capacity.

How a global external Application Load Balancer is assembled

1

Forwarding rule

One global anycast IP and port that clients connect to.

2

Target proxy

Terminates HTTP or HTTPS; holds SSL certificates (Google-managed or your own).

3

URL map

Routes requests by host and path to the right backend.

4

Backend service

Groups backends, defines balancing mode, health check, affinity and CDN.

5

Backends

Instance groups, network endpoint groups or Cloud Storage buckets.

EXAM LENS Traffic enters at the Google edge nearest the user and goes to the closest healthy backend with capacity.

Backends beyond instance groups

1 Backend buckets

Serve static content straight from Cloud Storage behind the same IP as your app.

2 Zonal NEGs

Point at containers or IP:port pairs, as GKE does for container-native load balancing.

3 Serverless NEGs

Front Cloud Run, App Engine or Cloud Run functions with a full load balancer.

4 Internet NEGs

Send traffic to an external origin, for example during a migration.

5 Hybrid NEGs

Include on-premises or other-cloud endpoints reached over VPN or Interconnect.

6 Session affinity

Keep a client on one backend using client IP or a generated cookie.

Cloud CDN caches content at Google's edge

Cache mode	Behaviour	Use when
CACHE_ALL_STATIC (default)	Caches static content types automatically; honours origin headers for the rest	General websites and mixed content
USE_ORIGIN_HEADERS	Caches only what the origin marks cacheable	Origins with precise Cache-Control headers
FORCE_CACHE_ALL	Caches every successful response, ignoring private directives	Purely public static content only

EXAM LENS Protect paid content with signed URLs or signed cookies; purge stale content with cache invalidation.

Proxy or passthrough at Layer 4

Proxy Network LB

Connections terminate at the proxy

- TCP, with optional SSL offload (global external)
- Global external mode uses one anycast IP worldwide
- Backends see the proxy's address unless PROXY protocol is on
- Internal and cross-region internal modes run on Envoy

Passthrough Network LB

Packets go straight to backends

- TCP, UDP, ESP, GRE, ICMP and ICMPv6 (external)
- Always regional
- Client source IP preserved; responses return directly
- Internal version can also be a next hop for routes

EXAM LENS UDP, or 'must see the client IP without headers' → passthrough.

Internal load balancing for private tiers

Internal load balancers expose a private IP so tiers inside the VPC can scale without being public.

Internal Application LB	Layer 7 routing between microservices; regional or cross-region, Envoy-based.
Internal proxy NLB	TCP proxy for private services; regional or cross-region.
Internal passthrough NLB	Fully distributed with no proxy; TCP or UDP; can front appliances.
Three-tier pattern	External ALB → web tier → internal LB → app tier → database.

Decision cheat sheet: picking the load balancer

If the scenario says…	Reach for
Global users, HTTP(S), one IP, path-based routing	Global external Application LB
Static assets fast worldwide	Backend bucket + Cloud CDN
Non-HTTP TCP, global, SSL offload	Global external proxy Network LB
UDP or preserve client IP, regional	External passthrough Network LB
Private HTTP microservices	Internal Application LB
Private TCP/UDP or route next hop	Internal passthrough Network LB
Serverless backend behind a load balancer	Serverless NEG
Scale on queue backlog	MIG autoscaling on a Monitoring metric



8 PCA-style scenarios

Read every requirement word, eliminate options that break one, then pick the most managed answer that satisfies them all. Each question is followed by its answer and the reasoning.

Question 1 of 8

Global HTTP

A media site serves users worldwide from one IP. Requests to /video must reach a GPU instance group in two regions, and /assets must be served from a Cloud Storage bucket.

A Global external Application Load Balancer with a URL map routing /video to a backend service and /assets to a backend bucket

B Two regional passthrough Network Load Balancers with DNS round robin

C Global external proxy Network Load Balancer with SSL offload

D Internal Application Load Balancer in each region

Answer 1: A

A Global external Application Load Balancer with a URL map routing /video to a backend service and /assets to a backend bucket

Why it fits

Path-based routing at Layer 7, one anycast IP and support for buckets as backends are all features of the global Application Load Balancer.

Why the others don't

B — No path routing and two IPs.

C — Layer 4 only; cannot route by URL path.

D — Not reachable from the internet.

EXAM LENS 'Path' or 'host' routing + global → Application LB with a URL map.

Question 2 of 8

UDP

A multiplayer game server uses UDP and needs to see each player's real source IP for anti-cheat checks. All servers run in one region.

A. Global external Application Load Balancer

B. External passthrough Network Load Balancer

C. Global external proxy Network Load Balancer

D. Cloud CDN

Answer 2: B

B External passthrough Network Load Balancer

Why it fits

The passthrough load balancer supports UDP and forwards packets without proxying, so backends see the original client IP. Regional scope fits.

Why the others don't

A — HTTP(S) only.

C — TCP only and terminates the connection.

D — A cache, not a UDP load balancer.

EXAM LENS UDP or 'preserve client IP' → passthrough NLB.

Question 3 of 8

Autohealing

A MIG keeps deleting and recreating VMs in a loop right after deployment. The application takes about 4 minutes to start, and the health check starts probing after 60 seconds.

A Switch to a stateful MIG

B Increase the autohealing initial delay to longer than the start-up time

C Disable the health check permanently

D Change the machine type to one with more memory

Answer 3: B

B Increase the autohealing initial delay to longer than the start-up time

Why it fits

The initial delay gives each VM time to boot before autohealing judges it. With 60 seconds, VMs are recreated before the app is ever ready.

Why the others don't

A — Preserves state but doesn't fix the timing.

C — Removes self-healing altogether.

D — Doesn't address the probe timing.

EXAM LENS 'Recreated in a loop after start' → initial delay too short.

Question 4 of 8

Health checks

After creating a new external Application Load Balancer, every backend shows UNHEALTHY, yet curl from inside the VPC to each VM returns 200 OK.

A. Enable Cloud CDN on the backend service

B. Add a firewall rule allowing ingress from 35.191.0.0/16 and 130.211.0.0/22 on the health-check port

C. Change the balancing mode to RATE

D. Recreate the instance template

Answer 4: B

B Add a firewall rule allowing ingress from 35.191.0.0/16 and 130.211.0.0/22 on the health-check port

Why it fits

Google's health-check probers come from those ranges. If the firewall blocks them, every backend appears unhealthy even though the app works.

Why the others don't

A — Caching doesn't affect health checks.

C — Balancing mode affects distribution, not health.

D — The VMs themselves are fine.

EXAM LENS 'Healthy from inside, unhealthy in LB' → probe ranges blocked.

Question 5 of 8

Internal tier

A web tier must call an internal API tier over HTTP, routing /orders and /billing to different backend services. The API must never be reachable from the internet.

A. Global external Application Load Balancer with Cloud Armor

B. Regional internal Application Load Balancer with a URL map

C. Internal passthrough Network Load Balancer

D. External passthrough Network Load Balancer with firewall rules

Answer 5: B

B Regional internal Application Load Balancer with a URL map

Why it fits

It provides a private IP plus Layer 7 path routing between services inside the VPC.

Why the others don't

A — Public entry point, even if filtered.

C — Layer 4; no path-based routing.

D — Externally exposed.

EXAM LENS Private + HTTP routing → internal Application LB.

Question 6 of 8

Autoscaling signal

Worker VMs pull jobs from a Pub/Sub subscription. CPU stays low while the backlog grows to millions of messages during peaks.

A. Lower the CPU target to 10%

B. Autoscale the MIG on a Cloud Monitoring metric: undelivered messages per instance

C. Use a stateful MIG

D. Schedule a fixed 100 VMs at all times

Answer 6: B

B Autoscale the MIG on a Cloud Monitoring metric: undelivered messages per instance

Why it fits

The backlog, not CPU, reflects demand. Scaling on undelivered messages per VM adds workers when work queues up and removes them when it drains.

Why the others don't

A — Over-provisions without tracking the real signal.

C — Statefulness is unrelated and disables autoscaling.

D — Wasteful outside peaks.

EXAM LENS Queue-driven workers → scale on backlog metric.

Question 7 of 8

Caching

Product images are slow for users in Asia and South America, and the origin bucket in us-central1 shows high egress. The images rarely change.

A. Copy the bucket to every region and use DNS geolocation

B. Enable Cloud CDN on the backend bucket behind the global Application Load Balancer

C. Move the images to Filestore

D. Increase the bucket's storage class to Standard

Answer 7: B

B Enable Cloud CDN on the backend bucket behind the global Application Load Balancer

Why it fits

Cloud CDN caches content at edge locations close to users, cutting latency and origin egress with a single configuration change.

Why the others don't

A — Operationally heavy and duplicates storage.

C — File storage isn't a content delivery solution.

D — Storage class doesn't change geographic latency.

EXAM LENS 'Static content' + 'global users' + 'slow' → Cloud CDN.

Question 8 of 8

TCP + TLS

A financial data feed uses a custom TCP protocol over TLS. Clients worldwide should connect to one IP, and you want TLS terminated before reaching backends in three regions.

A. Global external proxy Network Load Balancer with SSL offload

B. External passthrough Network Load Balancer

C. Global external Application Load Balancer

D. Internal proxy Network Load Balancer

Answer 8: A

A Global external proxy Network Load Balancer with SSL offload

Why it fits

It's global with one anycast IP, handles non-HTTP TCP and can terminate TLS at the edge before forwarding to the nearest healthy region.

Why the others don't

B — Regional and does not terminate TLS.

C — Expects HTTP(S), not a custom protocol.

D — Not internet-facing.

EXAM LENS Non-HTTP TCP + global + TLS offload → global proxy NLB.